

Solutions for Exercises Rigid Analytic Geometry

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Sheet 1

Problem 1

If $|\cdot|_K$ is ultrametric, then $K^{\circ\circ}$ is a unique maximal ideal. By the property of DVR, we have $x, y \in K^\circ$, $x \mid y$ or $y \mid x$. We can choose a minimal π among these ordering as $|\cdot|_K$ is discrete.

Problem 2

Recall that the norm $\|\cdot\|_{\mathbb{T}_m}$ is defined as,

$$f = \sum_{\alpha \in \mathbb{N}^m} f_\alpha X^\alpha \Rightarrow \|f\|_{\mathbb{T}_m} = \max_{\alpha \in \mathbb{N}^m} |f_\alpha|_K.$$

Let $f, g \in \mathbb{T}_m$. Then there are $\alpha, \beta \in \mathbb{N}^m$ such that

$$\|f\|_{\mathbb{T}_m} = |f_\alpha|_K, \|g\|_{\mathbb{T}_m} = |g_\beta|_K.$$

Then in the Taylor expansion of fg , $f_\alpha g_\beta$ arises as a part of the coefficient of $\gamma = \alpha + \beta$. By the ultrametric property of the absolute value, we see,

$$\|fg\|_{\mathbb{T}_m} \leq |f_\alpha|_K |g_\beta|_K = \|f\|_{\mathbb{T}_m} \|g\|_{\mathbb{T}_m}.$$

Obviously $\|1\|_{\mathbb{T}_m} = 1$.